Model Card: SafeSignal NER (Drug Side-Effect Extractor)

Detecting hidden drug side-effects from patient reviews using NLP

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1. Model details

Field	Description
Model name	SafeSignal-NER v1
Task	Named Entity Recognition: tag Drug, ADR, Disease, Symptom and Finding mentions in informal patient forum posts
Model	spaCy blank English pipeline (tok2vec + ner), trained from scratch for 20 epochs on CPU; training loss fell from 12,254 to 3,137
Training data	CADEC (CSIRO Adverse Drug Event Corpus): 1,250 AskaPatient.com posts about 12 drugs; 1,000 posts used for training
Test data	250 held-out posts (random 80/20 split, seed 42); entity-level precision, recall and F1
Preprocessing	BRAT annotations parsed, character spans converted to token spans (char_span), overlaps removed (filter_spans); about 4% of annotations skipped due to misaligned tokens; no lowercasing, stop-word removal or stemming
Libraries	Python, spaCy; Colab/Gradio demo app (the demo falls back to rule-based and demo engines; scores below are for the CADEC-trained model only)
Developed by	Sanobar Latif Shaikh (PRN 25030421039), SAII, Natural Language Processing course, October 2026

2. Intended use

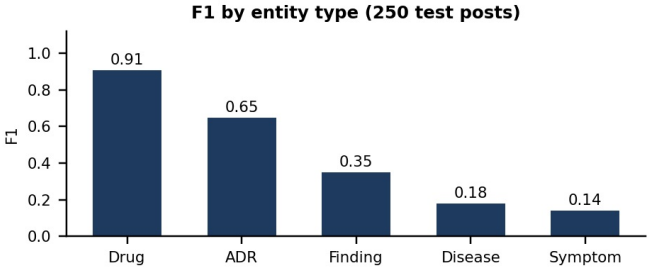
- **Intended:** Helping drug-safety reviewers spot possible adverse drug reactions in patient forum posts (early-warning triage); classroom demonstration of an NLP NER pipeline with an audit.
- **Out of scope:** Diagnosis or any decision about an individual patient; linking a reaction to the specific drug that caused it (no relation extraction); languages other than English and code-mixed text; fully automatic safety decisions without human review.

3. Overall performance

On the 250-post test set the model detects **Drug** very well (F1 0.91) and **ADR**, the most important class, reasonably (F1 0.65). Rare classes are weak (F1 0.14 to 0.35). Overall F1 is about 0.65 to 0.68 depending on text style (read from the audit chart).

Entity	Precision	Recall	F1
Drug	0.94	0.88	0.91
ADR	0.65	0.65	0.65
Finding	0.43	0.29	0.35
Disease	0.33	0.12	0.18
Symptom	0.50	0.08	0.14

Symptom: precision 0.50 but recall only 0.08, so the few tags are often right but most Symptoms are missed.



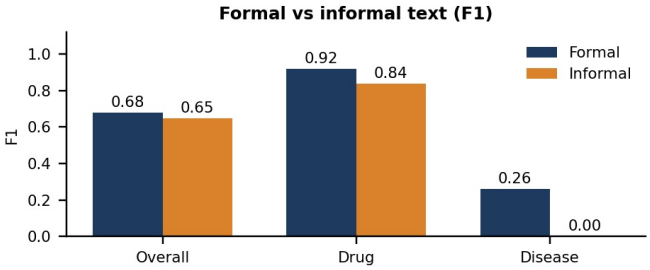
4. Performance across text subgroups

Key audit finding: Disease detection falls from F1 0.26 on formal text to 0.00 on informal text; Drug F1 drops by about 8 points (0.92 to 0.84) and overall F1 by about 3 points (0.68 to 0.65).

Subgroup	How it was defined	Formal F1	Informal F1
Overall	All entities	about 0.68	about 0.65
Drug	Drug entities only	about 0.92	about 0.84
Disease	Disease entities only	0.26	0.00

Formal = clearly written posts; informal = casual, misspelt or slang-heavy posts. Values marked "about" are read from the audit chart. Per-subgroup test sizes were not recorded, so small-sample uncertainty is not quantified.

The model learned mostly from clean annotated text, so it is least reliable on the casual language that real patients use.



5. Explainability check

Method 1: Per-class error pattern. Comparing precision and recall shows why rare classes fail: Disease (recall 0.12) and Symptom (recall 0.08) are missed far more often than they are wrongly tagged, which points to too few training examples, not noisy predictions.

Method 2: Span-level check on an unseen sentence. "I took Lipitor and started having severe muscle pain and memory problems." gave Lipitor = Drug, severe muscle pain = ADR, memory problems = ADR. The model separates the drug from the reactions even in casual wording, and the app shows these spans highlighted.

Not done: feature-attribution tools such as LIME or SHAP were not run, so token-level reasons for individual mistakes are not yet explained.

6. Limitations and ethical considerations

- **Main limitation:** Rare entity classes (Disease, Symptom, Finding) are unreliable because of class imbalance, and Disease detection fails completely on informal text. ADR output needs human review.
- **Data bias:** One forum, 12 drugs and English only, so the model reflects the people who use that site and may serve other groups worse, including casual writers.
- **Annotation loss:** About 4% of annotations were dropped because informal punctuation broke token boundaries, so some examples were never seen in training.
- **Privacy and safety:** Posts are personal health information; the tool is not medical advice and must support, not replace, human reviewers.

7. Recommendations

- Add more Disease, Symptom and Finding examples (or more data) to fix class imbalance.
- Try a biomedical transformer such as BioBERT and re-run the same formal vs informal audit to compare.
- Add spelling normalisation for informal text, and add relation extraction to link each ADR to its drug.
- Send low-confidence or rare-class predictions to a human reviewer.